University of Barishal

Department of CSE

3rd Year 2nd Semester; Mid term Exam; Session: 2022-23 (Admission: 2020-21)

Course name: Mathematical Analysis for Computer Science; Course code: CSE-3201

Total Marks: 20 **Time: 1.5 hour**

Answer all the questions. Marks are indicated at the end of the question

1. Define stochastic process with example. Describe different types of stochastic processes mentioning at least one appropriate example in each case. [4]

2. Consider the process $[X(t), t \in T]$ whose probability distribution, under a certain condition, is given by [4]

$$\Pr[X(t)=n]=\frac{(at)^{n-1}}{(1+at)^{n+1}} \quad , \qquad n=1,\ 2,\ \ldots\ \ldots$$

$$=\frac{at}{1+at} \quad , \qquad n=0$$

Test the stationarity.

3. What is martingale process? Let $\{Z_i\}$; $i=1,\ 2,\ \cdots$ be a sequence of *i.i.d.* random variables with $E\{Z_i\}=1$ [4]

and let $X_n=\prod_{i=1}^{n} Z_i$, then show that $\{X_n\}_{n=1}^{\infty}$ is a martingale.

4. Define random variable with an example. State down properties of random variables. [4]

5. Define stationary process and evolutionary process. Briefly explain Bernoulli process and Poisson process. [4]

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